

EXTRA! : correcting and creating meta-analyses with local language models — a pre-registered five-phase benchmark with a perturbation reading proof and a self-correcting answer key

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Abstract

Language models are entering systematic-review workflows, but most evaluations use cloud models, cannot separate reading from training-data recall, and treat the human answer key as infallible. EXTRA! is a pre-registered, five-phase benchmark of five quantized open-weight models (8–14.8B parameters, one consumer mini-PC, integrated GPU, no cloud calls) — plus a 27B low-bit build added afterwards as a pre-registered extension arm — as meta-analysis workers over two published meta-analyses and their 21 trials, under two controls: a *perturbation reading proof* (load-bearing numbers displaced under sealed maps, making recitation detectable) and a *two-layer, source-verified answer key* whose quotation-backed rite puts the human key itself on trial, documenting eleven source-confirmed discrepancies in one anchor and correcting its own cells twice. Reading: 168 perturbed sheets, zero attributable recitations, zero invention candidates, replicate stability 77–99%. Arithmetic: 1 of 73 unaided confidence intervals exact, rising to 86% through a trivial text-protocol calculator; pooling stayed out of reach for the 8–15B cast and was closed, 12 of 12, only by the 27B extension arm. Creation: *all six* models’ sheets reproduced the published dichotomous pools (morbidity 0.771–0.787 vs. 0.778), while the continuous diamond separated them — in the cast exactly one model, **gemma4:12b**, landed beside the published -0.24 , at -0.27 , the rest at -0.47 to -0.80 , and the extension arm landed closer still, at -0.25 [$-0.33, -0.16$] — never through arithmetic and never through fabrication — the invention screen found none. Orchestration: under a warn-only ten-net harness, confirm-or-correct resolved every warning and the model’s pooling call closed digit-consistent with its executed results. Deployment, on clean texts with no answer key: the model sided with the sources at all seven discrepancies reaching a single extractable cell, its one failed tolerance decomposed to one mislabeled-dispersion cell, and conversational and sheet-level nets proved complements, not substitutes. A language ablation (instruments drafted in Portuguese, frozen in English, identical materials) found every phase within replicate-level variance and the flagship diamond identical in both (-0.27 [$-0.38, -0.17$]): no conclusion is an artifact of instruction language. All protocols, seals, sheets and raw outputs are public.

1 Introduction

Systematic reviews and meta-analyses are the reference standard of clinical evidence synthesis, and they are labor-bound — registered reviews take a mean of 67 weeks from registration to publication [Borah et al., 2017] — extracting structured data from primary trials, checking it, and computing pooled effects consume the majority of a review team’s hours. Large language models are entering exactly this workflow. Feasibility studies report promising but uneven

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accuracy for trial data extraction [Schmidt et al., 2024]; assisted extraction has proved non-inferior to human-only extraction inside a live review [Lee et al., 2026]; screening assistance is approaching routine use [Xie et al., 2026]; and end-to-end platforms now report complete systematic reviews produced in hours [Bakker et al., 2026], with fully automated drafts rated above a published human-authored review by blinded specialists [McLaughlin et al., 2026]. The major evidence-synthesis organizations have jointly conditioned such use on demonstrated rigor, human oversight and transparent reporting [Flemyng et al., 2025].

Three gaps run through this literature. First, most evaluations use commercial cloud models, while clinical text pushes in the opposite direction: locally run open-weight models keep sensitive material on-premises and have shown extraction performance approaching expert annotation [Wiest et al., 2024, Spaanderman et al., 2025]. Second, benchmarks that score agreement with a published review cannot distinguish a model that *read* the primary trial from one that *remembers* the review — both the anchor and its primaries are plausibly in training corpora, and contamination inflates apparent competence [Xu et al., 2024]. Third, the published review itself is usually treated as an infallible answer key, although hand-made evidence tables demonstrably carry their own errors: checking 27 meta-analyses that used standardized mean differences against their primary trials, 17 (63%) contained an extraction error in at least one of the two trials examined, errors of a magnitude that can negate or reverse a finding [Gøtzsche et al., 2007], and a later methodological review found the evidence base for established extraction standards weak despite that high prevalence [Mathes et al., 2017]. A benchmark that cannot notice when the published value is the one that is wrong will misgrade its best readers.

EXTRA-I was designed to close those three gaps at once: quantized open-weight models on a consumer machine with no cloud calls; a perturbation reading proof, in which load-bearing numbers are displaced in every source text under sealed maps so that recitation of published values is detectable; and a two-layer, source-verified answer key with a quotation-backed adjudication rite that puts the human key itself on trial — a verification process that accumulated a seventeen-entry public record of source-verification findings on one published anchor — eleven confirmed discrepancies with the primary sources among them — while logging the adjudicators’ own mistakes in the same record [Barra, 2026]. Because the instruments were drafted in the research team’s working language before being frozen in English, the study additionally carries a built-in robustness check on instruction language, reported with the results.

In this study, we conducted a single pre-registered, five-phase benchmark of local language models as meta-analysis workers — reading, arithmetic, creation of two published meta-analyses, orchestration under a warn-only detection harness, and deployment on clean texts with no answer key in any loop — comparing five integrated-GPU models — and, in a pre-registered extension arm added after the campaign, a sixth — under contamination-controlled, source-verified grading against the published values.

2 Methods

2.1 Study design and pre-registration

This was a pre-registered, five-phase benchmark of five local language models as meta-analysis workers, graded against two published meta-analyses under a contamination control and a source-verified key. The study protocol — phase order, model cast, hypotheses and compute budget — was registered and committed to the public repository before any run; the two extensions executed within the deployment phase were likewise registered as dated amendments before their runs, as was the post-campaign extension arm that added a sixth model (Section 2.5). Corpora, sealed perturbations, answer keys, deterministic engines, graders and the detection

harness were constructed and frozen before the campaign, during an instrument-development phase whose complete record is archived alongside this study’s [Barra, 2026]; Section 2.13 describes the robustness analysis that phase additionally enables. All prompts, seals, keys, harness code, graders, raw model outputs, run logs and adjudication records are public.

The campaign comprised five phases in an ascending line of investigation: P1 *Read* (multi-model structured extraction under the perturbation reading proof), P2 *Calculate* (meta-analytic arithmetic unaided versus through a text-protocol calculator), P3 *Create* (deterministic engines assemble both meta-analyses from each model’s sheets, with a discrepancy-aware comparison against the published values), P4 *Orchestrate* (the best-performing reader executes its own calculations under a warn-only detection harness), and P5 *Deploy* (clean, unperturbed texts with no answer key in any loop, including a three-configuration comparison of error-detection architectures).

2.2 Anchor meta-analyses and corpora

Two recently published meta-analyses served as anchors. Anchor 1 addresses goal-directed fluid therapy (GDFT) versus conventional fluid therapy in major surgery [Ashraf et al., 2026]; its 14 primary randomized controlled trials constitute the first corpus (8 open-access, obtained from PubMed Central; 6 paywalled, legally obtained and never redistributed — scripts in the repository rebuild that stratum from a reader’s own copies). Anchor 2 addresses low-carbohydrate diets and glycemic control (HbA1c) in type 2 diabetes [Khraise, 2026]; its 7 primary trials constitute the second corpus. Anchor 1 supplies dichotomous outcome families (overall morbidity, mortality, postoperative ileus) and two continuous ones; as in any meta-analysis, each outcome pools only the trials that report it, so the anchor’s own outcome tables roster 4, 2 and 3 of the 14 trials for morbidity, mortality and ileus respectively — indeed, no outcome in the anchor is reported by more than 4 of its 14 trials, which meet all together only in its descriptive characteristics and risk-of-bias tables. All 14 trials are extracted here, each contributing to whichever outcome families it reports, and the extraction-level analyses (cell grading, stability, recitation and invention screens, and the per-trial three-way tables) span all 14. Anchor 2 supplies a single continuous outcome (change in HbA1c, %) reported by all 7 of its trials, whose published random-effects pooled estimate, -0.24 [$-0.32, -0.16$], is the study’s external reference value.

2.3 The perturbation reading proof

Before any model reads a trial, a sealed, deterministic operator displaces load-bearing numbers in its text. A model that returns a displaced value read the text in front of it; one that returns the original published value is reciting training data [Xu et al., 2024] — without this control, agreement with a published meta-analysis cannot be distinguished from memory of it. Models operate exclusively in the perturbed world; the original $\leftrightarrow$ perturbed maps are held by the graders alone and applied only after extraction, as a pure table lookup (the “unperturbation lens”). The sealed maps were produced once, before any model run, and reused unchanged across all phases; their SHA-256 fingerprints were printed into each phase’s run log. Phase P5 deliberately runs on clean texts and therefore makes no reading-proof claim (Section 2.11).

2.4 Answer keys and the adjudication rite

Each corpus is graded against a two-layer, source-verified answer key constructed during instrument development: layer 1 holds the anchor’s published cells; layer 2 holds each cell verified against the primary’s full text, with a literal supporting quotation and a per-cell verdict. The key itself is on trial by design — source verification of Anchor 1 documented a public, quotation-backed record of source-confirmed discrepancies in the published meta-analysis (Table 1; the

record also keeps the adjudicators’ withdrawn accusations, struck through, and its pending items), and the campaign graded against the key as corrected through the most recent adjudications. Anchor 2’s verification found no discrepancies: its statistics reproduced cleanly.

Table 1: Source-confirmed discrepancies between Anchor 1 and its primary sources, as graded against in this study (confirmed items only; full literal quotations, the withdrawn adjudicator accusations and pending items are in the archived verification record). Issuing formal errata is the journal’s and the authors’ prerogative; conclusion-level impacts are analyzed in the Discussion.

#	Where (anchor)	Published	Source says
1	Characteristics, Yoon	GDFT 36 / control 39	GDFT n=39; control n=36
3	Characteristics, Weinberg ASA	“Not stated”	I–II 7 (27%); $\geq$ III 19 (73%)
5	Characteristics, Sujatha ASA	“95:105” + a spreadsheet-time-corrupted cell	eligibility only (ASA I–II), no distribution
9	Oral-diet outcome, Sun	72 $\pm$ 24 vs 96 $\pm$ 30 h	“shorten ... by 2 days”; medians 4.0 vs 6.0 d
10	Characteristics, de Waal ASA	GDFT 24:123:86:1	inverse: 123 = 52.6% $\times$ 234 (control arm)
11	Characteristics, Diaper ASA	“Not stated”	III–IV 98 (50.0) / 85 (42.9)
12	Flatus outcome, 2 trials	55 $\pm$ 14/58 $\pm$ 16 and 52 $\pm$ 15/60 $\pm$ 18 h	“flatus” absent from both full texts
13	Sample sizes, six trials	presented as randomized	analyzed counts, undeclared
15	Pooled morbidity caption	“Mantel–Haenszel”	numbers reproduce under DerSimonian–Laird
16	Ileus outcome, Castro	ileus 6 (14.0%) / 19 (45.2%)	“ileus” absent; counts are the trial’s PPC result
17	Fluid table, Coeckelenbergh blood loss	GDFT 500 / control 450	the labeled parallel sentences give the inverse

Mechanical grading is backed by an adjudication rite: no cell changes category without the source quotation that decides it, and the adjudicators’ own errors are published in the same record as everyone else’s findings. The rite exists because judges exhibit a measurable *plausibility bias*: during key construction, the graded record reversed the benchmark’s own human–AI adjudicator six times — including two withdrawn accusations of model fabrication in which the models had extracted literally correct values that rigid verification windows had hidden [Barra, 2026]. Quotation-before-verdict is the structural constraint on that bias, for human, hybrid and model judges alike. The adjudication layer itself is that hybrid: a human–AI pair — the author working with a cloud-based frontier assistant that implements the graders and drafts verdicts under the rite (disclosed in the Declarations) — so the study’s local-only property is scoped to the *evaluated pipelines* (Section 2.6), never to the grading and review layer, and every adjudication this layer produced is recorded with its deciding quotation.

2.5 Instruments

All model-facing instruments are frozen English prompts, committed to the study’s record at registration: the structured extraction sheets (JSON forms with per-cell {"value", "where"} objects), the arithmetic tasks, the calculator protocol, the harness dialogue, and the trigger and synthesis prompts. The instruments were originally drafted in Brazilian Portuguese, the research team’s working language, and translated field-by-field under a published correspondence table (for example, `ic95_md`→`ci95_md`; `RESULTADO`→`RESULT`); the originals are archived, and Section 2.13 describes the robustness analysis this enables.

2.6 Models and hardware

Five quantized open-weight models sized for an integrated GPU formed the comparison cast: `gemma4:12b`, `qwen3:14b`, `llama3.1:8b`, `qwen3.5:9b` and `deepseek-r1:14b`, each pinned by its Ollama build digest in the repository’s model manifest. All phases ran on one consumer mini-PC (AMD Ryzen 7, Radeon 780M integrated GPU, 32 GB RAM, no discrete GPU), served by Ollama with a 16,384-token context, reasoning channels disabled, and a uniform 4,000-token output allowance for extraction tasks; no cloud API was called anywhere in any evaluated pipeline (the grading and adjudication layer, by contrast, is human–AI and cloud-assisted; Section 2.4). One model was resident at a time, with an explicit memory unload between models. Phase P4 used only `gemma4:12b`, pre-designated in the protocol as the orchestration candidate on instrument-development evidence.

After the campaign closed, a sixth model entered as a pre-registered extension arm (protocol Amendment 1, dated 2026-09-07 and committed before its run): `smttek/Qwen3.8-27B:Q2_K_XL`, an openly published mixed-precision quantization of a 27.3-billion-parameter model — the reader the author had meanwhile adopted as primary extractor in a separate systematic-review harness, and the reason for the arm. Its two printed names disagree and neither describes its weights (the tag reads `Q2_K_XL`; the GGUF `file_type` header, which is what the runtime prints, reads `Q4_K_S`), so the build was characterised by reading the tensor table of the weight file itself: 27.32 billion parameters in 10.68 GB at **3.13 bits per weight**, mixed across six ggml types (IQ3_XXS on 77.4% of parameters, IQ3_S on 11.7%, Q3_K and Q2_K on 4.7% each, IQ4_XS on 1.5%, IQ1_M on 0.1%). The cast’s five builds are Q4_K_M, about 4.5 bits per weight; resident footprints were 4.9–9.3 GB for the cast and 10.7 GB for the extension arm. It ran phases P1–P3 under the identical instruments, seals, keys, engines and graders, called exactly like the cast (same context, reasoning off, same output allowance, stock chat template), 54 calls in 119 minutes; P4 remained `gemma4:12b`’s by pre-designation. Its results are reported beside the cast’s and labelled as the extension arm throughout; after the graders were re-run over the extended cast, the five-model record was verified unchanged field by field.

2.7 Phase P1 — structured extraction

Each of the five models (and, later, the extension arm) extracted all 14 perturbed Anchor-1 primaries from zero, twice (two independent replicates), under the frozen English extraction sheet (30 fields per trial, each an object `{"value", "where"}` — the mandatory `where` is the model-side provenance instrument: a locator naming the section or table each value was taken from, not a verbatim quotation; these locators are raw data in every archived sheet and were load-bearing in adjudication, e.g. identifying which of three population layers a sample size came from). Grading used the eligible cell set of the two-layer key — cells with a usable, digit-bearing source-verified value; 124 cells per model — comparing the seal-reversed first-parseable sheet against the key’s source layer with a magnitude-based numeric comparator, declared an approximation of the full ruler (it cannot see summarization equivalence or population-layer choices); all non-matching cells were listed for adjudication rather than auto-verdicted. Replicate stability was the same comparator applied between the two raw replicates. Recitation candidates were detected mechanically — a cell whose raw text contains a seal pair’s original value while the displaced image the model actually read is absent — and each candidate was adjudicated against the perturbed source text before any verdict. Divergent cells were further classified by factuality: *omission* (the model wrote *not reported* where the source-verified key holds a value), *invention candidate* (the cell contains a number absent from the perturbed text and not derivable from it by percentage, sum, difference or mean), or *value/format divergence*; the same invention screen was applied to every cell of the Anchor-2 sheets of phase P3, both replicates. Beyond the three factuality classes, every non-matching cell was re-examined against the perturbed text its model read and classified by mechanism — equivalent re-encoding,

population-layer choice, case-mix summary, row- or scope-level slip, unmatched re-expression, or grader-side artifact — with residues adjudicated under the rite; the cell-by-cell record is Supplementary Tables S1–S7.

2.8 Phase P2 — meta-analytic arithmetic

Each model computed, over its own P1 sheets, per-study risk ratios and mean differences with 95% confidence intervals, and pooled estimates, in two arms: arm A unaided (no tool), and arm B through a plain-text calculator protocol in which the model writes lines of the form `CALC: function(arguments)`, the harness executes the validated function and returns `RESULT:` into the context (up to 5 rounds and 20 calls), and the model then answers with a final JSON. Three prompt families (risk ratios, mean differences, pools) $\times$ two arms $\times$ two replicates were run per model. Grading recomputed the ground truth with the same validated engine from the same input block the model saw, under pre-registered parsing rules (event counts read from forms such as “19 (32.8%)”; percent-only cells converted against the group size; means accepted only as mean $\pm$ SD, so that median (IQR) cells have no computable truth and *not computable* is the correct answer there). Each reported quantity was labelled exact, right-direction, wrong, correctly-refused, wrongly-refused, or unsupported; intervals and pooled estimates were graded separately.

2.9 Phase P3 — deterministic creation of both meta-analyses

Everything downstream of the sheets was executed by pre-registered, frozen code; no model touched a number after extraction. For Anchor 1, the engines parsed each model’s P1 sheets, selected event/denominator routes per outcome, computed per-study risk ratios and pooled them under both Mantel–Haenszel and DerSimonian–Laird [DerSimonian and Laird, 1986], applied the sealed reversal, and classified every difference from the published tables into a frozen five-category, discrepancy-aware comparison (*reproduces / differs by a documented anchor discrepancy / model route / model error / source unavailable*); the pooled comparison is made under DerSimonian–Laird because the published pooled numbers instantiate that method under a Mantel–Haenszel caption (documented discrepancy #15). For Anchor 2, each model freshly extracted the 7 perturbed primaries (two replicates) under the English continuous-outcome sheet — which, unlike the Anchor-1 sheet, records bare values with no provenance field, an instrument asymmetry inherited from its development version; a deterministic route selector then built each trial’s sextet (mean, SD, n per arm) by the frozen rules — declared change used directly; standard errors and confidence intervals converted to SDs; baseline-and-final pairs combined under the Cochrane $r = 0.5$ rule [Higgins et al., 2023] — and the DerSimonian–Laird pool was computed twice: in the perturbed world, and through the graders’ sealed lens for comparison with the published $-0.24 [-0.32, -0.16]$. Comparison tolerances were pre-registered at ± 0.01 for risk ratios and interval bounds and ± 0.1 for mean differences.

2.10 Phase P4 — orchestration under a warn-only detection harness

The study’s ten-net detection harness follows the doctrine that nets *detect and warn, never substitute a value*. The doctrine is empirical, not aesthetic: in instrument-development testing, checking stages given write-power did net harm through the same plausibility bias — an auditing model “repaired” correct cells by imputing plausible values (once correcting a sample size to 24.5 while quoting, in its own verdict, the text’s “ $n = 28$ ”), a calculator stage flipped the sign of a control arm that had genuinely worsened (+0.3 written as -0.3 , plausibility overriding the data it had just read), and a pipeline’s quality gate did six times more damage to the pooled estimate than the planted sabotage it caught [Barra, 2026]. The harness comprises: schema-constrained typed calls (`{"function", "arguments", "source", "derivation"}`), an arity check with

the full function signature, a sign-echo net, an arm-agnostic declared-source check, a declared-derivation check, an argument type system with per-class value impossibilities, a confidence-interval bound-order check, a coherence check tying each interval call to the study’s own last executed mean difference, a negative-SD result warning, a closing-versus-executed check, per-call warning budgets with confirm-by-repeat, and an escalation flag (`requires_human_review`) on insistence. `gemma4:12b` then ran one formal pipeline over its own Anchor-2 sheets: per-study typed calls, a model-emitted DerSimonian–Laird pooling call over its own sextets, a product-layer coherence check on the reported intervals (flagged, never endorsed), and a narrative synthesis checked mechanically for orphan numbers. No model stage saw an answer key, a published value, or the seal; grader-side comparison happened only afterward.

2.11 Phase P5 — deployment on clean texts

The deployment phase ran the same architecture on the *original, unperturbed* texts of both anchors with no answer key in any loop — the day-to-day usage scenario. On clean texts, agreement cannot distinguish reading from recall, so this phase makes no reading-proof claim — the perturbed phases carry that burden. Its components: (i) fresh two-replicate extraction of all 21 primaries by `gemma4:12b` under the English instruments; (ii) a frozen discrepancy-cell panel — the seven of Anchor 1’s eleven source-confirmed discrepancies that resolve to a single extractable cell with a decided source value, enumerated in the protocol before the run — testing whether the model’s cell sides with the source or with the anchor’s published value; (iii) deterministic recomputation of both meta-analyses and per-trial three-way tables (source quotation $\times$ the anchor authors’ cell $\times$ the model’s cell); and (iv) a three-configuration comparison of error detection over the same sheets — (a) the deterministic engine with no nets, (b) the engine plus deterministic sheet-versus-source nets, and (c) the model orchestrating under the conversational ten-net harness — each registered as a dated amendment before its run.

2.12 Comparison conventions and software

All computation used Python 3.12 with the repository’s validated engine functions (risk ratio and its log-scale interval; mean difference and its interval; SD conversions; Mantel–Haenszel and DerSimonian–Laird pooling; inverse-variance pooling), each frozen before the campaign; the risk-ratio function reproduces a published per-study value of Anchor 1 as its standing test case. No external statistics package was used. Replicates measure stability, not statistical significance, and no formal between-model inference was attempted: with one corpus per anchor the model comparisons are descriptive, and every superlative in this article is scoped to these corpora. The campaign’s compute totalled 277 model calls across P1–P4 (≈ 10.8 h of machine time), plus 54 calls (≈ 2.0 h) for the extension arm, with every queue resume-safe and every phase committed to the public repository before it ran.

2.13 Robustness analysis: instruction language

Because the instruments were drafted in Portuguese before being frozen in English, and the instrument-development phase had exercised the identical corpora, seals, keys, engines and graders under the Portuguese drafts, that archived record provides a natural comparison arm in which instruction language is the only changed variable. For each phase, the directly comparable quantities of the two records were compared, with the pre-registered acceptance criterion that differences stay within replicate-level variance.

3 Results

3.1 P1 — extraction: the reading proof held, and no attributable recitation occurred

All 140 extraction calls (5 models $\times$ 14 trials $\times$ 2 replicates) completed with clean stops in 390.4 minutes, and every sheet parsed at the sheet level (one single replicate, **deepseek-r1:14b** on one trial, was unparseable; its other replicate served). Cell agreement with the corrected two-layer key — after the four-cell adjudication of the caption — ranged from 89/124 (71.8%) for **llama3.1:8b** to 105/124 (84.7%) for **qwen3:14b** and **qwen3.5:9b**, with **gemma4:12b** at 103/124 (83.1%); replicate stability ranged from 77.4% to 99.2% (Table 2). The extension arm’s 28 sheets completed with clean stops in 77.9 minutes and all parsed; graded by the same comparator and the same one-cell adjudication, it scored 108/124 (87.1%) — the highest of the six — with replicate stability 117/124 (94.4%), six omissions and ten value/format cells (Table 2, last row). Because the numeric comparator cannot see format or granularity equivalence, cross-model ranking by this score alone is provisional; the value-level separation appears in P3. The point matters concretely and is visible in the table itself: **qwen3:14b**’s two-cell edge over **gemma4:12b** dissolves in the decomposition columns (six omissions against zero — and omissions are real misses where value/format variants are largely equivalence the comparator cannot see) and reverses decisively at the value level in P3.

The mechanical recitation detector raised 5 candidate cells across the cast’s 140 sheets (and none across the extension arm’s 28), and adjudication against the perturbed source texts resolved all 5 to incomplete-perturbation artifacts rather than recall: three models independently wrote a displaced count (113) that is arithmetically reconstructible from a surviving percentage printed beside it in the perturbed text ($57.7\% \times n=196 = 113.1$), and two models transcribed a value (4088) that survives verbatim in a prose restatement the perturbation operator missed. Zero attributable recitations remained. The same three models copied, rather than reconstructed, the equally inconsistent control-arm cell of the same table — an asymmetric preference shared across architectures, recorded as a behavior note.

The factuality classification of the divergent cells (Table 2, middle columns) found **zero invention candidates in every model**, the extension arm included: each number each model wrote exists in, or is arithmetically derivable from, the perturbed text it read. Most divergences were themselves replicate-stable — the same non-key answer written in both independent readings: 17/21 for **gemma4:12b**, 20/20 for **qwen3:14b**, 13/20 for **qwen3.5:9b**, 16/26 for **deepseek-r1:14b**, but only 17/36 for **llama3.1:8b**, and 16/17 for the extension arm (counted over the comparator’s raw divergent inventories, before the four-cell adjudication; **deepseek-r1:14b**’s denominator excludes one divergent cell in the trial whose second replicate was unparseable, leaving no pair to compare) — suggesting that for the stronger readers, divergence from the key is a systematic reading choice rather than sampling noise.

Supplementary Tables S1–S7 (after the references) give the complete cell-by-cell account of the 124 cells that failed the comparator across the five models — the divergences behind these counts, not a per-model total — each re-examined against the perturbed text its model read and classified by mechanism: 12 omissions; 27 equivalent re-encodings whose arithmetic verifies exactly (percentages for counts, rearranged count/percentage pairs); 26 population-layer choices (a different printed participant count of the same trial); 34 case-mix summaries; 12 genuine reading slips (an adjacent row, a transposed pair, a per-kilogram row, a different scope); 9 unmatched ratio-style re-expressions, all from **llama3.1:8b**; and 4 cells — one each for four models — adjudicated *in the models’ favor*: the transcription was verbatim-correct, and a seal-pair substring collision in the grader’s unperturbation lookup corrupted the comparison (Table S7). Table 2 carries the adjudicated scores; the comparator’s raw output stands in the archived record. The extension arm’s 17 divergent cells fall into the same mechanisms — the seven case-mix summaries every reader produces, the two FEDORA population-layer cells, one

re-encoding (0 (0) for 0 (0%)), six omissions and the same Castro collision cell, adjudicated in its favor by the same rule — and are enumerated in the repository’s Amendment-1 record rather than repeated in the supplement.

Table 2: P1 extraction under English instruments (perturbed Anchor-1 corpus; 124 eligible key cells per model). Two comparators: *stability* grades replicate 1 against replicate 2 (the model against itself, no key); *correct* grades replicate 1 against the source-verified key. The three middle columns decompose the non-matching cells by factuality (omissions + inventions + value/format = 124 – correct): *omission* = the model wrote *not reported* where the key holds a value; *invention* = a number absent from the source text and not derivable from it; *value/format* = content that traces to the source but differs from the key by value route (e.g. a different population layer) or by format of the same information (e.g. percentages for counts, a category summary for an enumeration). A cell can be stable yet divergent from the key, so stability can exceed the key score. Recitation candidates before/after adjudication. †one replicate unparseable (its pair served). One value/format cell each for **qwen3:14b**, **llama3.1:8b**, **qwen3.5:9b** and **deepseek-r1:14b** was adjudicated as a grader-side artifact — the transcription was verbatim-correct and a seal-pair substring collision in the unperturbation lookup corrupted the comparison (Supplementary Table S7); the four affected rows report the adjudicated scores (the comparator’s raw output was 104, 104, 97 and 88 correct, with value/format 14, 16, 25 and 36, preserved in the archived record). The last row, below the rule, is the post-campaign extension arm (Section 2.5), graded identically; its Castro collision cell was adjudicated by the same rule (raw 107 correct, value/format 11).

Model	Stability	Recit.	Omissions	Inventions	Value/format	Correct (vs key)
gemma4:12b	119/124 (96.0%)	1 → 0	0	0	21	103/124 (83.1%)
qwen3:14b	123/124 (99.2%)	1 → 0	6	0	13	105/124 (84.7%)
qwen3.5:9b	103/124 (83.1%)	0	4	0	15	105/124 (84.7%)
deepseek-r1:14b†	96/118 (81.4%)	1 → 0	2	0	24	98/124 (79.0%)
llama3.1:8b	96/124 (77.4%)	2 → 0	0	0	35	89/124 (71.8%)
qwen3.8:27b (ext.)	117/124 (94.4%)	0	6	0	10	108/124 (87.1%)

3.2 P2 — arithmetic: unaided fails, the calculator repairs

Across all five models, exactly 1 of 57 unaided 95% confidence intervals was exact (arm A). With the text-protocol calculator (arm B), exact intervals rose to 41/50 (82%) among the four models that operated the tool — 11/16 for **gemma4:12b**, 14/16 for **qwen3:14b**, 8/9 for **llama3.1:8b**, 8/9 for **qwen3.5:9b** — and exact point estimates rose in step (4→12, 4→14, 5→8, 5→8). Pooled estimates remained at or near zero exact in both arms for every model in the cast, with or without the tool. **deepseek-r1:14b**’s unaided arm was unusable (5 of 6 outputs unparseable, zero gradable quantities), and its pooled-family tool arm ran without emitting a single **CALC:** line. The extension arm reproduced the unaided signature (0/16 exact intervals, 6 exact points) and, with the calculator, closed every quantity its sheets support: 16/16 exact points, 16/16 exact intervals and — uniquely among the six — 12/12 exact pooled estimates, with no malformed output.

3.3 P3 — creation: every model reproduces the dichotomous pools; one of the cast, and the extension arm, reach the continuous diamond

From the P1 sheets, the deterministic engines reproduced Anchor 1’s published pooled estimates for *all five* models and for the extension arm: pooled morbidity 0.771–0.787 against the published 0.778 [0.567, 1.068], and pooled mortality 0.980–1.023 against the published 1.021 [0.446, 2.337], every value within ± 0.01 of the published except one mortality estimate at $\Delta 0.04$ (Table 3). The extension arm’s morbidity pool sits farthest from the cast’s (0.787) for an instructive reason: on FEDORA it read the literal event counts (18/224 and 35/226) and reproduced the published per-study risk ratio 0.519 exactly, where **gemma4:12b** read the percentages and derived 19/38 — a route difference the ratio structure again dampened to $\Delta 0.009$ at the pool.

The postoperative-ileus pool is non-comparable by construction (documented discrepancy #16: one published row is a different outcome) and was reported, never pooled against: over the two genuinely-ileus trials, the pool was 0.821 [0.559, 1.207] under Mantel–Haenszel and 0.429 [0.048, 3.87] under DerSimonian–Laird — both crossing 1 — identically from the perturbed and the clean-text extractions.

Table 3: P3, Anchor 1: DerSimonian–Laird pools from each model’s own sheets (published: morbidity 0.778 [0.567, 1.068]; mortality 1.021 [0.446, 2.337]). Every model contributed every trial of the anchor’s own per-outcome roster (4/4 for morbidity, 2/2 for mortality — a meta-analysis pools, per outcome, only the trials reporting it; Section 2.2), so no trials-in-pool column is needed here — unlike Anchor 2 (Table 4), where sheet starvation varies by model.

Model	Morbidity RR [95% CI]	Mortality RR [95% CI]
gemma4:12b	0.779 [0.569, 1.065]	1.023 [0.447, 2.344]
qwen3:14b	0.779 [0.569, 1.065]	1.021 [0.446, 2.337]
llama3.1:8b	0.771 [0.562, 1.059]	1.023 [0.447, 2.344]
qwen3.5:9b	0.779 [0.569, 1.065]	1.019 [0.445, 2.336]
deepseek-r1:14b	0.775 [0.565, 1.063]	0.980 [0.428, 2.246]
qwen3.8:27b (ext.)	0.787 [0.580, 1.068]	1.023 [0.446, 2.346]

From the fresh Anchor-2 extractions (70/70 calls, 155.1 minutes; the extension arm’s 14/14 in 27.6 minutes), the same engine and sealed lens produced the cast’s five diamonds and the extension arm’s sixth (Table 4). **gemma4:12b**’s lens landed at -0.27 [-0.38 , -0.17] (I^2 28.6%), beside the published -0.24 [-0.32 , -0.16]; the extension arm’s landed at -0.25 [-0.33 , -0.16] (I^2 7.1%, τ^2 0.001), the closest of the six — both fall within the pre-registered ± 0.1 tolerance at the point estimate and at both bounds, and the extension arm’s reconstructs the anchor’s own homogeneity (published I^2 6%). The other four landed between -0.47 and -0.80 with saturated heterogeneity (I^2 90.5–99.1%), two of them with starved sheets (5/7 and 6/7 trials in the pool). Reversal itself separated the readers the same way: it collapsed **gemma4:12b**’s pool heterogeneity from I^2 91.3% in the displaced world to 28.6% and the extension arm’s from 91.2% to 7.1%, and left the other four models’ I^2 unchanged to within 0.1 points (Table 4). The invention screen over all 1,104 Anchor-2 sheet cells (five models, both replicates) found zero invention candidates in any model; a re-implementation of the same rule over the extension arm’s 224 Anchor-2 cells (196 numbers) found none either, and returned zero for the five as well.

Table 4: P3, Anchor 2: heterogeneity anatomy of the cast’s five diamonds, and the extension arm’s, against the published (human) extraction — pooled HbA1c mean differences through the unperturbation lens, i.e. each model’s sheet with the graders’ sealed number-map reversed by pure table lookup before pooling (Section 2.3); models extracted in the displaced world and never saw the originals, and the lens corrects nothing — reading errors pass through it intact. The published row is as the anchor prints it (I^2 6%, τ^2 0.00). The last column is the same pool *before* reversal, in the displaced world; it demonstrates the correct-nothing property mechanically — reversal collapses **gemma4:12b**’s I^2 from 91.3% to 28.6% (in faithful sheets the perturbation displacement *was* most of the heterogeneity) while leaving the other four models’ I^2 unchanged to within 0.1 points (heterogeneity injected by reading survives the lookup by construction).

Extraction	Trials in pool	Lens MD [95% CI]	τ^2	I^2	I^2 before reversal
Published (human)	7/7	−0.24 [−0.32, −0.16]	0.00	6%	—
qwen3.8:27b (ext.)	7/7	−0.25 [−0.33, −0.16]	0.001	7.1%	91.2%
gemma4:12b	7/7	−0.27 [−0.38, −0.17]	0.005	28.6%	91.3%
deepseek-r1:14b	7/7	−0.47 [−1.22, 0.28]	0.997	99.1%	99.1%
llama3.1:8b	5/7	−0.60 [−1.02, −0.18]	0.199	90.5%	90.4%
qwen3:14b	6/7	−0.63 [−1.08, −0.17]	0.291	95.2%	95.2%
qwen3.5:9b	6/7	−0.80 [−1.67, 0.07]	0.898	95.0%	95.1%

3.4 P4 — orchestration under the English harness

gemma4:12b closed all 7 per-study call sequences under the English ten-net harness in 8.7 minutes. Two content warnings fired — both on one trial, both from the declared-derivation net — and both were resolved by the confirm-or-correct mechanism; no other net fired and no format failure occurred. The model then emitted the pooling call over its own seven sextets and closed digit-consistently with the executed result (−0.50 [−0.78, −0.22]); the mechanical truth over the same sheets was −0.52 [−0.82, −0.22] (Δ MD 0.02, from two named route divergences). The product layer flagged zero incoherent intervals, and the synthesis contained zero orphan numbers. The grader-side lens over the same sheets was −0.27 [−0.38, −0.17].

3.5 P5 — deployment on clean texts: the discrepancies are visible, and the one miss is one cell

On the original, unperturbed texts, replicate agreement was 96.8% (Anchor 1) and 95.4% (Anchor 2), and all 42 sheets parsed. At every cell of the frozen discrepancy panel — 7 of 7, including all four direction-critical swaps — the model’s extraction sided with the *source* against the anchor’s published value; on the trial whose published ileus row reproduces its pulmonary-complication counts (discrepancy #16), the model wrote *not reported* in both replicates rather than reciting the published numbers. The deterministic recomputation reproduced Anchor 1’s pooled morbidity (0.779 [0.569, 1.065]) and mortality (1.019 [0.445, 2.336]) within tolerance; the Anchor-2 pool missed its tolerance at −0.34 [−0.51, −0.18] against the published −0.24 [−0.32, −0.16], and the residue decomposed entirely to one cell — the model transcribed a printed “−1.6 ± 0.3” from one trial’s results prose and declared the ± a standard deviation, where the value equals the half-width of the confidence interval printed in the same trial’s table. Correcting that single cell by the key’s own conversion rule yields −0.24 [−0.32, −0.16] (I^2 3.7%), the published diamond to the hundredth; the cell was identical in both replicates, and zero invented values occurred in the phase’s 124 graded cells.

Over the same sheets, the three detection configurations behaved differently against that poisoned cell (Table 5): the bare engine delivered the distorted pool silently; the sheet-versus-source net flagged the experimental arm mechanically, with both source quotations, after a first rule revision (its initial form stayed correctly silent because the trial’s own prose prints the ± form — the flag rule was extended, as a dated amendment, to fire when a printed ± equals the

printed interval’s half-width; the control arm’s $\pm$ matches nothing within the pre-registered tolerance and was not flagged); and the conversational ten-net harness raised zero flags on the poisoned cell — whose downstream calls were well-formed, correctly source-declared and internally coherent — while catching four live orchestration slips (one arity, three derivation declarations) in the same run. A DerSimonian–Laird weight-share flag ($>40\%$) was blind to the distortion (the poisoned trial’s share was 19.3%; the distortion’s product signature was heterogeneity inflation, I^2 6% $\rightarrow$ 79%).

Table 5: P5: three error-detection configurations against the same single-cell defect, same sheets, no answer key in any loop.

Configuration	Pool delivered	The poisoned cell
(a) deterministic engine, no nets	−0.34 [−0.51, −0.18]	silent
(b) engine + sheet-vs-source nets	−0.34 (unchanged; warn-only)	flagged (exp. arm, with quotes)
(c) model + conversational ten-net harness	−0.32 [−0.46, −0.19]	not flagged; 4 live slips caught

3.6 Instruction-language robustness

Per phase, the campaign’s results sat within replicate-level variance of the archived instrument-development record, in which identical materials had run under the Portuguese instrument drafts (Table 6). The largest reading-stage difference was +3 cells of 124 on the one directly comparable model, with replicate stability identical to the decimal; the arithmetic signature (unaided failure, calculator repair, pool ceiling, per-model deviant classes) matched point for point; the creation-stage flagship lens was identical in point estimate *and* interval (−0.27 [−0.38, −0.17] in both records); and the orchestration-stage delta sat inside the measured route-variability band with the same warning classes on the same trial.

Table 6: Instruction-language robustness: the campaign (English instruments) vs the archived instrument-development record (Portuguese drafts), per phase (directly comparable quantities only).

Phase	Campaign (English)	Development record (Portuguese)
Read (gemma4:12b, cells / stability)	83.1% / 96.0%	80.6% / 96.0%
Calculate (unaided exact CIs; tool arm)	1/57; 82% exact	0/30; same signature
Create (gemma4:12b lens, Anchor 2)	−0.27 [−0.38, −0.17]	−0.27 [−0.38, −0.17]
Orchestrate (pool vs truth; behavior)	Δ 0.02; confirm-or-correct	Δ 0.06; same classes

4 Discussion

This study measured, in one continuous pre-registered chain, what small local models running on an integrated GPU can contribute to meta-analysis — and where the risk concentrates. The models read clinical trials reliably enough to rebuild two published meta-analyses; they cannot do the interval arithmetic unaided but do it almost perfectly through a trivial text-protocol tool; a deterministic architecture turns their sheets into pooled estimates that match the published ones; and, within the pre-registered cast, exactly one model produces a continuous-outcome diamond beside the published value — a 27B low-bit extension arm added afterwards reproduced it more closely still. Under the perturbation control, none of this is attributable to memory of the published review.

The six-model comparison exposes a clean division of difficulty, and the mechanism appears to be structural as much as cognitive. On the dichotomous side, the raw material is transcription — integer counts and denominators printed redundantly across table, prose and percentage — and the pipeline then *dampens* what error remains: a risk ratio is a ratio, so choosing a different population layer shifts numerator and denominator together, and the pooling weights derive from the counts themselves. The evidence is in Table 3: the six models took visibly different denominator routes and still landed at 0.771–0.787 against the published 0.778. On the continuous side each sextet quantity is printed once, in one of several competing layers, and the decisive step is semantic rather than transcriptive — identifying what a printed dispersion *is*; a mislabeled type sends the conversion down the wrong route, and inverse-variance pooling then *amplifies* the error, because weight scales with $1/\text{SD}^2$ [Higgins et al., 2023] — which is how one dispersion four-fold too small dominated a pool and inflated I^2 from 6% to 79% in the deployment phase. The same architecture is therefore error-dampening for count outcomes and error-amplifying for continuous ones. This extends the feasibility literature’s central observation — extraction accuracy that is promising but uneven [Schmidt et al., 2024] — by giving the unevenness a structure: in this record it tracks the outcome type at least as much as the model, and it is the synthesis architecture downstream of the sheet, not the reader alone, that decides whether a reading error reaches the pooled estimate.

A related question deserves a direct answer: nothing corrects a divergent cell anywhere in the chain — the downstream is deterministic, the nets only warn, the graders only classify, and the unperturbation lens is a pure table lookup that maps displaced numbers back without touching errors (the four failing Anchor-2 diamonds passed through the same lens and stayed far) — Table 4’s before-reversal column shows the same neutrality mechanically: reversal collapsed only the displacement-injected heterogeneity of the faithful sheets (I^2 91.3% to 28.6%), while the reading-injected heterogeneity of the other four passed through unchanged. The pooled values match the published ones despite the divergent cells of Table 2 because those cells concentrate in baseline descriptor fields the engines never consume — the pooled quantities draw only on the event and denominator cells, which were correct — and in population-layer choices the ratio structure dampens. The six diamonds spread from -0.25 to -0.80 entirely through reading-class differences, never arithmetic — the same engine computed all six — supporting the study’s central design thesis: once everything deterministic is code, reading is the whole game, and model choice is a reading choice. The extension arm sharpens what *well-chosen* means: at a comparable memory footprint (10.7 GB against 4.9–9.3 GB for the cast) it carried roughly twice the parameters at about seven-tenths the precision — 27.3 billion weights at 3.13 bits against 8–14.8 billion at 4.5 — and read better than every model in the cast on both anchors. In this record the parameter count, not the bit depth, tracked reading quality; with one build per model the two cannot be separated, and the observation is a lead for the next campaign rather than a result of this one. Even the cast’s best diamond, however, is not a full reproduction of the anchor’s homogeneity: **gemma4:12b**’s lens carries I^2 28.6% against the published 6% — the same pooled effect, direction and significance over a visibly noisier reconstructed evidence body, with the residual heterogeneity plausibly injected by route-level extraction differences (population layers, dispersion conversions) rather than by any single wrong cell; the extension arm’s 7.1% shows that homogeneity to be reachable by reading alone. “Beside the published value” is therefore a claim about the estimate, not digit-level equivalence of the whole meta-analysis — an honest bound worth stating, since the wider interval is exactly what a practitioner would inherit.

Notably, none of those reading failures was a fabrication: across both corpora and all six models, the invention screen found no cell whose numbers cannot be traced to the text read, and the failure inventory consists of omissions, population-layer choices, format variants and mislabeled quantities — most of them reproduced identically across replicates in the stronger readers, i.e. systematic and therefore auditable, not stochastic. This honesty profile is what

makes a warn-only architecture viable at all — detection nets suffice for models whose errors are traceable, whereas fabricating models would demand the value-substituting interventions the doctrine forbids. The profile also agrees with what a cloud-model extraction study reported independently: across 160 data items its errors were few and *missed data was the most common error type* [Gartlehner et al., 2024]. That two evaluations at opposite ends of the model-size and deployment spectrum find omission dominating fabrication suggests the failure shape is a property of the extraction task rather than of one model class — and it is the shape that decides which safeguards are needed. The pooled-arithmetic ceiling points the same way: even with the calculator, assembling the list-valued pooling calls stayed beyond every model of the 8–15B cast, while the 27B extension arm closed all twelve — consistent with the instrument-development observation that only a 27-billion-parameter model ever mastered that call — which is precisely why the architecture moves pooling into code rather than relying on the reader’s size. For practitioners this suggests a concrete triage: dichotomous evidence tables may be delegated broadly, while continuous outcomes concentrate all the model risk and deserve the strongest reader plus the strongest checks.

The deployment phase turns the benchmark toward real use, where no answer key exists. Its clean-text results support the day-to-day scenario in three ways. First, the model’s extractions sided with the primary sources at all seven of the eleven source-confirmed discrepancies that resolve to a single extractable cell with a decided source value — the remaining four being a method-caption error, a sample-size convention spanning six trials, and two cells whose sources report nothing verifiable to side with (Table 1) — including refusing, in both replicates, to supply the ileus counts that the anchor publishes but its source never reports — so a local reader plus a three-way table (source quotation, human cell, model cell) makes published extraction discrepancies *visible* to a reader rather than requiring trust in either extractor. One of them carries conclusion-level weight (Table 1, #16): the anchor’s significant ileus reduction (abstract RR 0.48) rests on the mislabeled pulmonary-complication row — over the two genuinely-ileus trials the pool crosses 1 under either method (0.821 [0.559, 1.207] MH; 0.429 [0.048, 3.87] DL) — while its no-difference conclusions for morbidity and mortality survive every documented discrepancy, as the five-model reproductions demonstrate. This is consistent with reports that language-model assistance can match human extraction quality in live reviews [Lee et al., 2026], and it adds the complementary observation that the disagreements are informative in both directions: the same source-verification process also corrected the benchmark’s own key twice during development, and the public record carries the adjudicators’ own published mistakes. Second, the phase’s one failed tolerance decomposed to a single cell with an instructive anatomy: the primary itself prints the same quantity in two layers — a table interval and a prose “ $\pm$ ” that equals the interval’s half-width — and the model transcribed the prose layer and mislabeled the $\pm$ as a standard deviation, identically in both replicates. The defect is seeded by the source’s own ambiguous notation, and it defines this architecture’s characteristic residual failure class: a model that misdeclares its sheet escapes every check that trusts the sheet. This failure class is not peculiar to machine readers, which is why it deserves the dedicated net rather than a better model: an audit of a published sports-medicine meta-analysis found the same dispersion mix-up in 9 of its 16 effect sizes, and correcting it moved the pooled estimate from a standardized mean difference of 1.46 [0.91, 2.01] to 0.54 [0.37, 0.71] and cost a second comparison its significance, over a trial literature in which nearly one report in five describes variability with a standard error rather than a standard deviation [Sandercock, 2024]. Our single poisoned cell distorted a pool in exactly that direction and by the same mechanism, which suggests the sheet-versus-source dispersion net earns its place in any pipeline, human or model. The instrument asymmetry likely mattered here: the continuous sheet carries no provenance field, so the sheet could not say *which* printed layer the 0.3 came from — on the dichotomous sheet, whose mandatory **where** locator repeatedly disambiguated population layers during adjudication, the prose-versus-table conflict would have surfaced at extraction; extending provenance (or short

verbatim-quote) fields to every sheet is the corresponding instrument work, queued for future versions and never retroactive. Third, the three-configuration comparison shows that detection layers are complements, not substitutes: conversational nets catch conversation-layer defects (four live orchestration slips in a single run) but are structurally blind to a well-formed poisoned cell, which only a sheet-versus-source net saw — and a weight-share product flag was blind too, because random-effects weighting re-equalizes shares while the distortion surfaces as heterogeneity inflation. The orchestration phase supplies the complementary case, and the two phases together delimit what a warn-only net can accomplish. Under the same harness in P4, two declared-derivation warnings fired across seven per-study sequences and the confirm-or-correct mechanism resolved both — the net states what it saw, the model confirms or revises, and no value is ever overwritten — after which the model’s own pooling call closed digit-consistent with its executed results (Δ MD 0.02 against the mechanical truth over the same sheets). A net that can only point still changes outcomes, provided a model is there to answer it. No warn-only configuration made the wrong number right — by doctrine it must not — so the safest key-free architecture this record supports is a deterministic engine, sheet-versus-source detection nets, a conversational harness for orchestration slips, and a reviewer dispatched to the primary’s table whenever any flag fires — a role played in this study by the human–AI adjudication pair, whose flag resolutions are themselves quotation-bound and on record.

These results sit within a fast-moving literature. Feasibility studies report promising but uneven extraction accuracy for cloud models [Schmidt et al., 2024]; screening assistance is approaching routine [Xie et al., 2026]; and end-to-end platforms report complete reviews in hours [Bakker et al., 2026, McLaughlin et al., 2026]. In contrast to that line, everything measured here runs locally, at zero marginal cost, on hardware compatible with the privacy constraints of clinical text [Wiest et al., 2024, Spaanderman et al., 2025] — and under evidence-synthesis organizations’ stated conditions of demonstrated rigor and transparent reporting [Fleming et al., 2025], which the perturbation proof, the public seals and the adjudication rite are designed to operationalize. The reading-versus-recall control matters more, not less, as anchors and primaries enter training corpora [Xu et al., 2024]: in 168 perturbed sheets (the extension arm included) the campaign found zero attributable recitations, while both mechanical candidates it did raise traced to survivable gaps of the perturbation operator itself — now demonstrated across multiple architectures and queued as instrument fixes.

These discrepancies are also a measurement of the human side of extraction, and the two extractor kinds fail differently — with one telling overlap. That the human column carries errors at all is consistent with the methodological literature rather than remarkable within it: 63% of a sample of published meta-analyses that used standardized mean differences contained an extraction error in at least one of the two trials audited [Götzsche et al., 2007], and the standards meant to prevent this remain weakly evidenced [Mathes et al., 2017]. What this study adds to that line is the source-quotation trail cell by cell, and a model column audited beside the human one. The anchor’s eleven source-confirmed discrepancies (Table 1) are transcription and labeling slips: swapped arms, values published as “not stated” that the primaries state, a conversion contradicting its own source, a method caption naming the wrong estimator, an outcome published under the wrong heading — and, like the models, the human extractors fabricated nothing. The models’ profile concentrates instead in omissions, layer choices and format variants. The overlap is the instructive part: the one human error with conclusion-level weight and the one model error that distorted a pool are the *same class* — a mislabel (an outcome heading; a dispersion type) — suggesting that label semantics, rather than numbers, is where extraction fails for both kinds of extractor, and therefore where verification effort should concentrate. We note the fairness symmetry the benchmark imposes on this comparison: the human errors were found by the same quotation rite that graded the models and that published the adjudicators’ own mistakes.

A thread that runs through both the instruments and the findings is plausibility bias —

the tendency of any judge, human, hybrid or model, to prefer a plausible reading over the printed one. It shaped this study twice before a single campaign run: the adjudication rite exists because the benchmark’s own judge was reversed six times during key construction, twice while wrongly accusing models of fabrication, and the warn-only doctrine exists because development-phase checkers with write-power repaired correct values into wrong ones. In the campaign itself the bias was then measured, not presumed — and it appeared exactly once, on the judge’s side: the deployment phase’s record initially charged the model with *computing* the mislabeled dispersion, until a detection-net run showed the value printed verbatim in the source’s own prose, and the charge was corrected with a dated note in the archived record [Barra, 2026]. The same containment later caught the grading layer a second time, in a different failure class — mechanical, not judgmental: four byte-perfect transcriptions of one trial’s blood-loss cell were charged as divergences because the reversal lookup applied a two-digit seal pair inside an unrelated longer number; the cells were adjudicated in the models’ favor and Table 2 carries the corrected scores (Supplementary Table S7). The model stages showed zero occurrences: no value was ever substituted (structurally forbidden), the sign-echo net never fired, and all six models transcribed even the sign-sensitive cells verbatim — while their overall factuality profile ran opposite to the judges’ (zero fabrications; failures traceable to the text). The published meta-analysis’s errors were caught by quotation rather than judgment, and the three-configuration comparison rewards architectures whose checks can only point, never overwrite. This aligns with the documented biases of model judges [Zheng et al., 2023] and suggests they generalize beyond model judges: the constraint that works is structural — quote before verdict, flag before fix — not a better judge.

The instruction-language robustness check closes a confounder rather than opening a finding. Because the instruments were drafted in Portuguese and frozen in English over otherwise identical materials, a referee could ask whether either language handicapped — or peculiarly helped — models trained predominantly on English; the comparison answers by measurement. Per phase the two records agree within replicate-level variance — the one directly comparable reading score moved three cells in 124, the arithmetic signature matched point for point, and the flagship diamond was -0.27 $[-0.38, -0.17]$ in both, interval included — which suggests that multilingual instruction-following is effectively solved at these model sizes for structured tasks, and establishes that none of the conclusions above is an artifact of instruction language.

Several limitations bound these claims. Two anchors from a single journal family and 21 primaries are the whole evidence base; every superlative is scoped to these corpora, and no formal between-model inference is attempted. One quantization per model and one hardware profile were tested, so precision and parameter count are not separable here: the extension arm was added after the campaign, run once, and not taken through P4, and its advantage is measured against five builds that differ from it in both. The mechanical cell comparator is a declared approximation of the full adjudicated ruler, so absolute cell scores understate agreement and cross-model gaps partly reflect format habits. The perturbation operator’s residual gaps (derivable totals; prose restatements) are documented and exploitable, though they produce reconstruction, not recall. The English harness build reproduces the frozen net logic but was exercised in one formal run of one model. The adjudicator is the same human–AI pair that built the harness, mitigated by mandatory quotation and public self-correction rather than independence, and the known biases of model judges [Zheng et al., 2023] are constrained, not eliminated, by that rite. Finally, the deployment phase’s clean texts cannot separate reading from recall by design; its claims rest on the perturbed phases’ proof.

In conclusion, a consumer machine running one well-chosen local reader — a 12-billion-parameter model at 4.5 bits per weight, or, at the same per-call latency and a comparable footprint, 27 billion parameters at 3.13 — a frozen deterministic engine and warn-only detection nets can rebuild published meta-analyses to the published digits, expose documented errors in them with source quotations, and fail in ways that are named, single-celled and flaggable — and

none of this depends on the instruction language. The designed next step is the generalization gate: the same frozen architecture against a new anchor in a new clinical domain, where transfer, not replication, is the question.

Data and code availability

All protocols (pre-registered, with dated amendments committed before each run), the frozen English instrument library with its Portuguese-correspondence tables, perturbation seals with SHA-256 fingerprints, the two-layer answer keys and the anchor source-verification record, the deterministic engines and graders, the English harness build, all 331 raw model outputs and run logs (277 from the campaign, 54 from the extension arm), per-phase evaluation records, adjudications with source quotations — consolidated in a raw-data index of the adjudication layer (`dados/adjudication-record.md`: the quotation-bearing key cells, the judge’s reversals and withdrawn accusations, the campaign-era correction with its refuting net run, and every flag resolution) — and the exact quantized builds (pinned by Ollama digest) are public at <https://github.com/gbbarra/llm-evidence-extraction-benchmark-ptbr> and archived at Zenodo [Barra, 2026]. Closed-stratum primary articles are not redistributed; scripts rebuild that corpus stratum from legally obtained copies.

Declarations

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Competing interests. The author declares no competing interests. For transparency: the AI assistant declared below (Claude, Anthropic) was used under an ordinary consumer subscription paid by the author, and the author has no financial or professional relationship with Anthropic or with the organizations whose open-weight models are evaluated here.

Use of artificial intelligence. This campaign was designed and conducted by the author with the assistance of Claude (Anthropic), which translated and froze the English instruments, implemented the phase runners, the English harness build and the graders (all committed before their runs), performed the mechanical adjudications under the quotation rite, and drafted this manuscript under the author’s direction and review. The evaluated local models produced only the extraction sheets, the arm-A/B arithmetic answers and the phase-P4 harness calls graded here; every downstream number in this article was produced by the frozen code and verified mechanically. Per the position of the evidence-synthesis organizations [Flemyng et al., 2025], every AI contribution is reported and the author retains responsibility for the work.

Ethics approval and consent. Not applicable. This study involved no human participants and no new data collection from humans: it analyzes only published, aggregate-level reports of randomized controlled trials and two published meta-analyses. No institutional review board approval or participant consent was required.

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Supplementary material: P1’s 124 comparator failures across the five models, cell by cell

This supplement enumerates every non-matching cell of the READ phase (Table 2): all 124 cells, across the five models, where a sheet value — after seal reversal — failed the numeric comparator against the source-verified key. Each cell was re-examined against the perturbed source text its model actually read, and classified by mechanism; the class totals are exhaustive ($12+27+26+34+12+9+4 = 124$). Model abbreviations: G = `gemma4:12b`, Q14 = `qwen3:14b`, L = `llama3.1:8b`, Q35 = `qwen3.5:9b`, D = `deepseek-r1:14b`; a parenthesized “(2)” after a field means both arms’ cells. The reading of the whole: no cell is an invention — every value traces to the text read; 87 cells (Tables S2–S4) are faithful to the source in a different encoding, population layer or resolution; 12 cells (Table S5) are genuine reading slips — a real value taken from the wrong row, arm or scope; 9 cells (Table S6) are ratio-style re-expressions matching no printed layer, all from the cast’s weakest reader; and 4 cells (Table S7) were adjudicated in the models’ favor as a grader-side artifact. The extension arm’s 17 divergent cells (Section 3.1) are enumerated, under the same classes, in the repository’s Amendment-1 record (`dados/estudo8/avaliacao-amend1.md`).

Table S1: Omissions (12 cells): the sheet says *not reported* where the source prints a value. The Diaper rows carry a nuance the class name hides: the source prints only a partial datum (the ASA III–IV share), and writing NR for a full-distribution field is defensible conservatism; the key transcribes the partial datum.

Trial	Field	Source value (the key)	NR written by
Sun	inotrope use	GDFT 18 (36%) vs. control 27 (54%)	Q14, Q35
Wu	inotrope use	norepinephrine 15 (25.9%) vs. 24 (42.9%); phenylephrine 12 (20.7%) vs. 24 (42.9%); ephedrine 12 (24.0%) vs. 19 (35.2%)	Q14, Q35
Redondo	ASA (2)	0:10:6:0 (control); 0:13:6:0 (GDFT)	Q14, D
Calvo			
Diaper	ASA (2)	only the ASA III–IV share is printed: 85 (42.9%) vs. 98 (50.0%)	Q14, Q35

Table S2: Equivalent re-encodings (27 cells): the same printed datum re-expressed in another notation; in every row the arithmetic was verified against the key’s counts (e.g. Castro control ASA II: $26/42 = 61.9\% \approx 62$). The comparator, which matches digits, cannot see these equivalences — the limitation Table 2’s caption declares.

Trial	Field	Key (source layer)	What was written
Castro	ASA (2)	11:26:5:0 / 7:31:5:0	counts as percentages, 26/62/12 and 16/72/12 — as % strings (G), as a ratio string (L); all match the counts over the arm totals
de Waal	ASA (2)	24:123:86:1 / 17:132:95:4	as percentages 10.3/52.6/36.7/0.4 and 6.9/53.2/38.3/1.6 (Q35, D); exact over the analysis-set totals
FEDORA	ASA (2)	35:136:40:0 / 14:122:73:0	as percentages 16.6/64.4/19.0 and 6.7/58.4/34.9 (D); exact
Wu	ASA	0:22:34:0 (control); 0:25:33:0 (GDFT)	control as percentages 39.3/60.7 (L, D); GDFT as percentages 43.1/56.9 (D); exact
FEDORA	laparoscopy (2)	109 / 105	as percentages 51.7% (= 109/211) and 50.2% (= 105/209) (D)
Yoon	laparoscopy (2)	0 (open surgery)	“0%” (D) — same meaning
Sujatha	laparoscopy (2)	0 (open surgery)	“excluded” (G) — same meaning, no digits for the comparator
Arslan-Carlon	postoperative ileus (2)	30 (21.3%) / 36 (25.4%)	rearranged with denominator and rounded: “21% (30/141)”, “25% (36/142)” (G, L, Q35, D) — counts identical after seal reversal; the raw sheets show the perturbed image (31) that reversal maps back to 36, i.e. faithful perturbed-world reading

Table S3: Population-layer choices (26 cells): the model filled an *n randomized* cell with a different participant count printed by the same trial. The key anchors each cell on the source-verified layer of the anchor’s own cell, so the direction varies by trial — for de Waal the key holds the randomized counts and models wrote analysis-population numbers; for FEDORA and Hokenek the key (like the anchor) carries the analysis set and models wrote the allocation, the field name’s literal layer. The dampening mechanism discussed with Table 3 is why none of this moved a pooled estimate: ratio outcomes shift numerator and denominator together.

Trial	Field	Key (source layer)	What was written
de Waal	n randomized (2)	274 / 259 (randomized)	printed analysis populations: 258/244, the complications analysis (“PGDT group (n = 258)” ... “control group (n = 244)”) (G, L); 248/234, the hypotension analysis denominators (Q14); and mixtures reusing the control arm’s 259 (Q35, D)
FEDORA	n randomized (2)	211 / 209 (analysis set)	226 / 224 — the allocation: “450 patients were randomized to the GDHT group (n=224) or control group (n=226)” (G, Q14, L, Q35)
Hokenek	n randomized (2)	39 / 39 (analysis set)	40 / 40 — the allocation: “(control group: 40, PVI group: 40)” (G, Q14, L)
Sujatha	n randomized, control	102 randomized	101 — the analysed count, the layer the anchor’s own cell carries (L)
Diaper	n randomized, control	201	200 — the other arm’s allocation; the power calculation also prints “200 patients in each group” (L)

Table S4: Case-mix summaries (34 cells): the *surgery type* field condensed to a category label where the key transcribes the source’s enumerated case mix. Faithful but lossy — no wrong fact, less resolution than the key’s reading.

Trials	Models	Example
Yoon, Sun, Sujatha, Diaper, de Waal, all five FEDORA		Yoon — key: “Eight open radical cystectomies with ileal conduit, 31 open...”; written: “open radical cystectomy”
Schmid	G, Q14, L, Q35	key: “16 Whipple, 71 oesophageal, three pancreatectomies...”; written: “major non-cardiac surgery” (G), “major abdominal surgery” (L)

Table S5: Different row or scope picked (12 cells): genuine reading slips — the value exists in the source, in the wrong place. These, not fabrications, are the READ phase’s real failure inventory.

Trial	Field	Key (source layer)	What was written
Castro	blood loss (2)	the “Intraoperative bleed-ing” row: 1283.2 ± 959.7 / 1100.1 ± 851.1	the same table’s “Losses” row: 5062.9 ± 3287.5 / 4364.5 ± 2920.6 (G)
Castro	colloid (2)	Voluven (mL): 737.8 ± 383.2 / 779.7 ± 565.1	the adjacent per-kilogram row, Voluven (mL kg^{-1}): 10.7 ± 5.9 / 10.8 ± 6.6 (L)
Sun	ASA (2)	8:33:9:0 / 6:36:8:0	adjacent rows mixed: control “16 (32%)” is the table’s Female row; the GDFT ASA-I cell carries the control arm’s 8 (D)
Wu	ASA, GDFT	0:25:33:0 (= 43.1/56.9%)	the two shares transposed: 56.9/43.1 (L)
FEDORA	morbidity, control	35 (16.6%)	36 — a nearby printed value (L)
Schmid	ASA, GDFT	NR (source prints eligibility criteria only)	a distribution filled from another scope (L, D)
Arslan-Carlon	ASA, GDFT	NR (no per-arm distribution in the text)	the overall “30% II / 70% III–IV” placed in the per-arm field (D); placeholder junk “I:II:III:IV (NR:NR:NR:NR)” (L)

Table S6: Unmatched re-expressions (9 cells, all `llama3.1:8b`): ratio-style strings that match no printed layer as a set. The digits are individually present in or derivable from the text read — which is why the invention screen does not fire — but the assembled distributions correspond to nothing the source prints; sheet residue, held for adjudication. The same model is also the cast’s least replicate-stable (17/36 of its divergents stable), so part of this residue may be sampling noise rather than a systematic reading.

Trial (ASA field)	Key (source layer)	<code>llama3.1:8b</code> wrote
Yoon (2)	2:27:7:0 / 4:29:6:0	0.75:0.24:0.01 / 0.74:0.25:0.01
de Waal (2)	24:123:86:1 / 17:132:95:4	1:2.9:3.2:0.6 / 1:2.8:3.2:1.2
Diaper (2)	only the III–IV share is printed	full I–IV ratios 47:38:22:13 / 50:50:30:10
Redondo Calvo (2)	0:10:6:0 / 0:13:6:0	class-shifted 10:4:2:0 / 13:3:0:0
FEDORA, GDFT	14:122:73:0	partial shift: I:122, II:148, III:73, IV:0

Table S7: Grader-side lookup collision (4 cells): the models were right. Adjudicated with the source open, in the same quotation-bound rite the key itself is subject to; Table 2 reports the adjudicated scores, and the comparator’s raw output is preserved in the archived record. The same mechanics that caught the human anchor’s discrepancies caught the grader here.

Trial	Field	What happened
Castro	blood loss, control	Q14, L, Q35 and D transcribed the printed “Intraoperative bleeding (mL) 1283.2 ± 959.7 ” verbatim — byte-identical to the key’s source value. The unperturbation lookup then applied the trial’s seal pair $31 \rightarrow 28$ (an ASA count) as a substring replacement <i>inside</i> “ <u>1283.2</u> ”, yielding “1313.2” and failing the comparison. Correct transcription; comparator artifact. (G alone genuinely diverged on this cell — the “Losses” row; Table S5.)